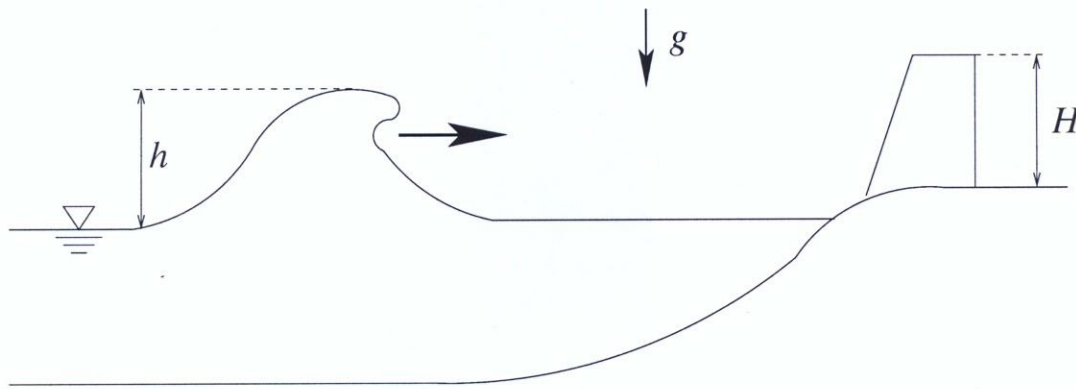


A protection wall against *tsunamis* is to be built at the coast. An experiment is to be conducted to evaluate the impact force that the tsunami, assumed to be of height h , may exert on the wall, of height H . To guide the design and analysis of the experiment, follow these steps:

- Use the Π theorem to simplify the parametric dependence of the *tsunami* velocity U . (2 points)
- Use the Π theorem to simplify the parametric dependence of the impact force **per unit width of wall** F . (2 points)
- Simplify the above relationships assuming that the viscosity has a negligible effect. Indicate the condition that needs to be satisfied. (1 point)

The real wall has a height $H = 3$ m and is to be designed to withstand *tsunamis* of height $h = 2$ m. If the laboratory model has a height $H' = 0.3$ m:

- Determine the laboratory value of the tsunami height h' necessary to guarantee physical similarity. (2 points)
- For that value, the laboratory measurements provide $F' = 1500$ N/m and $U' = 2$ m/s. Determine the values of F and U for the real tsunami. (2 points)
- Verify the hypotheses of negligible viscosity for the model and for the real tsunami. (1 point)



$$① \quad U = U(s, g, h, \mu) \rightarrow \frac{U}{\sqrt{gh}} = f_1\left(\frac{\mu}{\rho g^{1/2} h^{3/2}}\right)$$

$$② \quad F = F(\rho, g, h, H, \mu) \rightarrow \frac{F}{\rho g h^2} = f_2\left(\frac{\mu}{\rho g^{1/2} h^{3/2}}, \frac{H}{h}\right)$$

$$③ \quad \text{IF } \frac{\mu}{\rho g^{1/2} h^{3/2}} \ll 1 \rightarrow \frac{U}{\sqrt{gh}} = f_1 = \text{const}, \quad \frac{F}{\rho g h^2} = f_2\left(\frac{H}{h}\right)$$

$$④ \quad \frac{H}{h} = \frac{H'}{h'} \Rightarrow \boxed{h' = \frac{h}{H} H' = 0.2 \text{ m}}$$

$$⑤ \quad U/\sqrt{gh} = U'/\sqrt{gh'} \Rightarrow U = \sqrt{\frac{h}{h'}} U' = 6.32 \text{ m/s}$$

$$\frac{F}{\rho g h^2} = \frac{F'}{\rho g h'^2} \Rightarrow F = \left(\frac{h}{h'}\right)^2 F' = 150,000 \text{ N/m}$$

$$⑥ \quad \frac{\mu}{\rho g^{1/2} h^{3/2}} \approx 10^{-7} \ll 1$$

$$\frac{\mu}{\rho g^{1/2} h'^{3/2}} \approx 3 \times 10^{-6} \ll 1$$